Supplementary file 7 - Predictors of medication (non-)adherence

Study (country)	Population	Predictors assessed	Dependent variable	Significant predictors of and/or associations with medication adherence
Beadles 2015 (USA) [1]	≥ 2 of the following: major depressive disorder, schizophrenia, diabetes mellitus, hypertension, hyperlipidaemia, seizure disorder, asthma, chronic obstructive pulmonary disease	• Medical home enrolment	Proportion Days Covered (PDC) ≥ 0.8	• Medical home enrolment was positively associated with PDC ($p < .001$) across all cohorts (see original paper for breakdown by cohort)
Cicolini 2016 (Italy) [2]	≥ 2 of the following: cardiovascular, neurological, respiratory, musculoskeletal, oncologic diseases or diabetes mellitus	• Age • Sex • Education • Marital status • Number of prescribed drugs • Presence of diabetes • Presence of other disorders (cancer, musculoskeletal, respiratory, neurological, other) • Beliefs about medications (necessity, concerns) • Beliefs about medications (ambivalent, accepting)	8-item Moriskey Medication Adherence Scale (MMAS-8); "medium adherence" vs "low adherence"	• Necessity beliefs: OR (95% CI) = 1.61 (1.21-2.14), $p < .001$ • Concern beliefs: OR (95% CI) = 2.02 (1.64-2.49), $p < .001$ • Ambivalent vs accepting: OR (95% CI) = 0.24 (0.16-0.37), $p < .001$
Daniels 2016 (USA) [3]	≥ 2 chronic conditions	• Health literacy • Medication assistance • Medication adherence self-efficacy • Health-related problem solving • Functional health literacy • Chronic disease self-efficacy • Quality of life • Number of chronic conditions • Number of medications	MMAS-8; higher score = lower adherence	• Health-related problem solving (total score): $r = -.489$, $p = 0.024$) • Health-related problem solving (impulsive subscale): $r = -.521$, $p = 0.0015$ • Quality of life (physical subscale): $r = -.44$, $p < .05$ • Quality of life (environment subscale): $r = -.566$, $p = 0.009$

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Domino 2014 (USA) [4]	≥ 2 of the following: asthma, COPD, diabetes, hypertension, hyperlipidaemia, seizure disorder, depression, schizophrenia	Adherence in medical index condition: • Number of comorbid medical conditions • Mental health conditions Adherence in mental health condition: • Number of comorbidities	PDC (results show percentage point change (standard error) in adherence associated with clinical diagnosis stated)	Persons with diabetes: • Depression: -0.0427 (0.0030), $p < .01$ • Schizophrenia: 0.0458 (0.0061), $p < .01$ • 2 medical comorbidities: 0.0201 (0.0037), $p < .01$ • ≥ 3 medical comorbidities: 0.0678 (0.0037), $p < .01$ Persons with hyperlipidaemia: • Depression: -0.0200 (0.0025), $p < .01$ • Schizophrenia: 0.0359 (0.0050), $p < .01$ • 2 medical comorbidities: 0.0447 (0.0037), $p < .01$ • ≥ 3 medical comorbidities: 0.0979 (0.0037), $p < .01$ Persons with hypertension: • Depression: -0.0276 (0.0021), $p < .01$ • Schizophrenia: -0.0105 (0.0041), $p < .05$ • 2 medical comorbidities: 0.0365 (0.0023), $p < .01$ • ≥ 3 medical comorbidities: 0.1039 (0.0024), $p < .01$ Persons with depression: • 1 medical comorbidity: 0.0193 (0.0029), $p < .01$ • 2 medical comorbidities: 0.0462 (0.0035), $p < 0.01$ • ≥ 3 medical comorbidities: 0.0894 (0.0037), $p < .01$ Persons with schizophrenia: • 1 medical comorbidity: 0.0362 (0.0067), $p < .01$ • 2 medical comorbidities: 0.0752 (0.0078), $p < 0.01$ • ≥ 3 medical comorbidities: 0.1082 (0.0079), $p < .01$
Eton 2017 (USA) [5]	≥ 2 chronic conditions requiring self-management	• Health provider relational quality (HPRQ)	Single-item self-report; “always” vs “usually/sometimes” taking medications	• Health provider relational quality: t (294) = 3.79, $p < .001$ (converted to $r = 0.22$, $p < .001$ for consistency with other studies)

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Ihle 2017 (Switzerland) [6]	≥ 2 chronic conditions	• Prospective memory • Retrospective memory	Omission errors of adherence = 1-item self-report Commission errors of adherence = 1-item self-report	• Omission errors of adherence: prospective memory performance: r = -.29, p < .05 • Commission errors of adherence: retrospective memory performance: r = -.24, p < .05
Kenning 2015 (UK) [7]	Any combination of the following: chronic obstructive pulmonary disease, coronary heart disease, diabetes, osteoarthritis and depression.	• Gender • Age • Deprivation (IMD) • Number of conditions • Health status • Anxiety/depression (HADS) • Illness perceptions – consequences (IPQ) • Illness perceptions – treatment control (IPQ) • Self-help behaviours • Self-monitoring and insight (heiQ) • Baseline Moriskey medication motivation • Baseline Moriskey medication knowledge • Hassles with healthcare • Patient experience of multimorbidity (MULTIPleS)	Modified Moriskey Scale – Medication Motivation at follow-up	• Health status (B = -0.119, SE = 0.051; β= -0.132, p < .05) • Self-monitoring and insight (B = 0.093, SE = 0.045; β = 0.100, p < .05) • Moriskey medication motivation at baseline (B = 0.522, SE = 0.044; β = 0.552, p < .01). • Hassles with health care (B = -0.107, SE = 0.050; β = -0.108, p < .05)
Schüz 2011 (Germany) [8]	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Unintentional non-adherence Time 1 (T1) • Intentional non-adherence T1 • Sex • Age • Education • Number of illnesses • Number of medicines • Change in specific necessity (T1 to T2) • Change in specific concerns (T1 to T2) • Change in general harm (T1 to T2) • Change in general overuse (T1 to T2)	Two items from Reported Adherence to Medication Scale; Higher score = higher non-adherence.	Intentional non-adherence as dependent variable: • Intentional non-adherence time 1: B = .58, SE = .09, p < .01 • Change in specific necessity beliefs: B = –.19, SE = .06, p < .01 Unintentional non-adherence as dependent variable: • Unintentional non-adherence time 1: B = .70, SE = .09, p < .01 • Change in general overuse beliefs: B step 3 = .26, SE = .10, p < .01

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Schüz 2011 (Germany) [9]	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Unintentional non-adherence Time 1 • Intentional non-adherence Time 1 • Sex • Age • Education • Functional health status • Number of medicines • Number of illnesses • General harm beliefs • General overuse beliefs • Specific necessity beliefs • Specific concern beliefs	Two items from Reported Adherence to Medication Scale; Higher score = higher non-adherence	Unintentional non-adherence as dependent variable: • Unintentional non-adherence time 1: β step 3 = .29, $p < 0.01$ • General overuse beliefs: β step 3 = .17, $p < .01$ Intentional non-adherence as dependent variable: • Intentional non-adherence time 1: β = .26, $p < .01$ • Number of medicines: β = -.15, $p < .05$ • Number of illnesses: β = .16, $p < .05$ • General harm beliefs: β = .15, $p < .05$ • Specific necessity beliefs: β = -.15, $p < .05$
Schüz 2014 (Germany) [10]	≥ 2 chronic conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Illness perceptions (illness consequences, timeline, control)	One item from Reported Adherence to Medication Scale. Higher score = higher adherence.	Main effects and interaction model: • Timeline of illness 2: β = -.18, $p < .05$ • Control of illness 2: β = .21, $p < .05$ Peak model: • No significant predictors Combined model: • Higher control: β = .20, $p < .01$ • More negative consequences: β = .16, $p < .05$
Warner 2013 (Germany) [11]	≥ 2 chronic physical conditions from Charlson Comorbidity Index or Functional Comorbidity Index	• Sex • Age • Education • Number of medicines • Physical fitness • Attitudes towards medication • General received support • Social conflict • Medication specific social support Time 1 • Medication specific social support Time 2 • Medication adherence Time 1 • Medication adherence Time 2	One item from Reported Adherence to Medication Scale assessing unintentional non-adherence. Higher score = higher non-adherence	Step 1: OR (95% CI) • Adherence Time 1 = 5.99 (3.18-11.29), $p < .001$ Step 2: OR (95% CI) • Adherence Time 1 = 5.37 (2.82-10.24), $p < .001$ • Medication specific social support = 0.69 (0.49-0.98), $p < .05$. Step 3: OR (95% CI) • Adherence Time 1 = 5.62 (2.91-10.85), $p < .001$ • Interaction medication support x social conflict = 0.42 (0.17-0.99), $p < .05$

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